Exercises: XML Processing

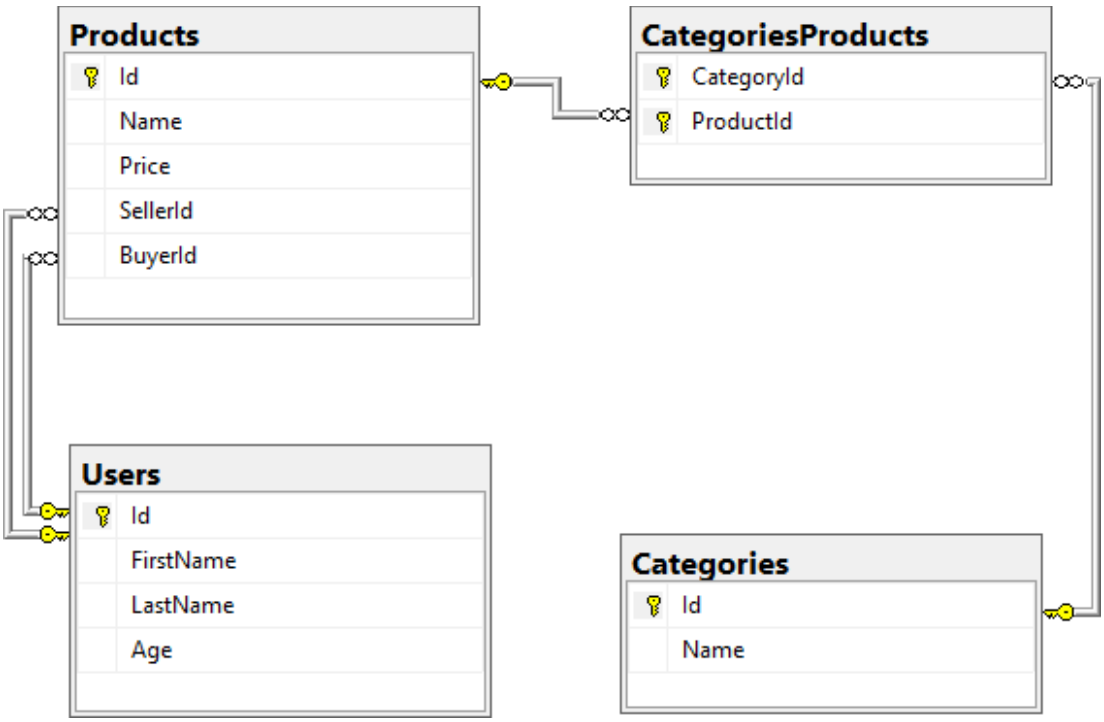
You can check your solutions in Judge

Product Shop Database

1. Setup Database

A products shop holds **users**, **products** and **categories** for the products. Users can **sell** and **buy** products.

- **Users** have an **Id**, **FirstName** (optional), **LastName** and **Age** (optional).
- **Products** have an **Id**, **Name**, **Price**, **BuyerId** (optional) and **SellerId** as **Id**s of **Users**.
- **Categories** have an **Id** and **Name**.
- Using Entity Framework and Code First create a database, following the above description.



- **Users** should have **many Products sold** and **many Products bought**.
- **Products** should have **many Categories.**
- **Categories** should have **many Products.**
- **CategoryProducts** should **map Products** and **Categories.**

2. Import Data

Query 1. Import Users

NOTE: You will need method **public static string ImportUsers(ProductShopContext context, string inputXml)** and **public StartUp** class.

Import the users from the provided file "**users.xml**".

Your method should return a string with the following message:
`$"Successfully imported {users.Count}";`

Query 2. Import Products

NOTE: You will need method **`public static string ImportProducts(ProductShopContext context, string inputXml)`** and **`public StartUp`** class.

Import the products from the provided file "**`products.xml`**".

Your method should return a string with the following message:
`$"Successfully imported {products.Count}";`

Query 3. Import Categories

NOTE: You will need method **`public static string ImportCategories(ProductShopContext context, string inputXml)`** and **`public StartUp`** class.

Import the categories from the provided file "**`categories.xml`**".

Some of the names will be null, so you don't have to add them to the database. Just skip the record and continue.

Your method should return a string with the following message:
`$"Successfully imported {categories.Count}";`

Query 4. Import Categories and Products

NOTE: You will need method **`public static string ImportCategoryProducts(ProductShopContext context, string inputXml)`** and **`public StartUp`** class.

Import the categories and products ids from the provided file "**`categories-products.xml`**". If provided **`CategoryId`** or **`ProductId`** doesn't exist, skip the whole entry!

Your method should return a string with the message:
`$"Successfully imported {categoryProducts.Count}";`

3. Query and Export Data

Write the below-described queries and **export** the returned data to the specified **format**. Make sure that Entity Framework Core generates only a **single query** for each task.

Query 5. Export Products In Range

NOTE: You will need method **`public static string GetProductsInRange(ProductShopContext context)`** and **`public StartUp`** class.

Get all products in a specified **price range** between **500** and **1000** (inclusive). Order them by price (from lowest to highest). Select only the **product name**, **price** and the **full name of the buyer**. Take top **10** records.

Return the list of suppliers **to XML** in the format provided below.

products-in-range.xml

```
<?xml version="1.0" encoding="utf-16"?>
<Products>
  <Product>
```

```
    <name>TRAMADOL HYDROCHLORIDE</name>
    <price>516.48</price>
    <buyer> </buyer>
  </Product>
  <Product>
    <name>Allopurinol</name>
    <price>518.5</price>
    <buyer>Wallas Duffyn</buyer>
  </Product>
  <Product>
    <name>Parsley</name>
    <price>519.06</price>
    <buyer>Brendin Predohl</buyer>
  </Product>
...
</Products>
```

Query 6. Export Sold Products

NOTE: You will need method **public static string GetSoldProducts(ProductShopContext context)** and **public StartUp** class.

Get all users who have **at least 1 sold item**. Order them by **the last name**, then by **the first name**. Select the person's **first** and **last name**. For each of the **sold products**, select the product's **name** and **price**. Take top **5** records.

Return the list of suppliers **to XML** in the format provided below.

users-sold-products.xml

```
<?xml version="1.0" encoding="utf-16"?>
<Users>
  <User>
    <firstName>Almire</firstName>
    <lastName>Ainslee</lastName>
    <soldProducts>
      <Product>
        <name>Ampicillin</name>
        <price>674.63</price>
      </Product>
      <Product>
        <name>Strattera</name>
        <price>658.54</price>
      </Product>
      <Product>
        <name>Aspergillus repens</name>
        <price>1231.42</price>
      </Product>
    </soldProducts>
  </User>
...
</Users>
```

Query 7. Export Categories By Products Count

NOTE: You will need method **public static string GetCategoriesByProductsCount(ProductShopContext context)** and **public StartUp** class.

Get **all categories**. For each category select its **name**, the **number of products**, the **average price of those products** and the **total revenue** (total price sum) of those products (regardless if they have a buyer or not). Order them by the **number of products** (**descending**), then by total revenue (**ascending**).

Return the list of suppliers **to XML** in the format provided below.

categories-by-products.xml

```
<?xml version="1.0" encoding="utf-16"?>
<Categories>
  <Category>
    <name>Garden</name>
    <count>23</count>
    <averagePrice>800.150869</averagePrice>
    <totalRevenue>18403.47</totalRevenue>
  </Category>
  <Category>
    <name>Weapons</name>
    <count>22</count>
    <averagePrice>671.073181</averagePrice>
    <totalRevenue>14763.61</totalRevenue>
  </Category>
...
</Categories>
```

Query 8. Export Users and Products

NOTE: You will need method **`public static string GetUsersWithProducts(ProductShopContext context)`** and **`public StartUp`** class.

Select users who have **at least 1 sold product**. Order them by the **number of sold products** (from highest to lowest). Select only their **first** and **last name**, **age, count** of sold products and for each product - **name** and **price** sorted by price (descending). Take top **10** records.

Follow the format below to better understand how to structure your data.

Return the list of suppliers **to XML** in the format provided below.

users-and-products.xml

```
<Users>
  <count>54</count>
  <users>
    <User>
      <firstName>Dale</firstName>
      <lastName>Galbreath</lastName>
      <age>31</age>
      <SoldProducts>
        <count>9</count>
        <products>
          <Product>
            <name>Fair Foundation SPF 15</name>
            <price>1394.24</price>
          </Product>
          <Product>
            <name>Finasteride</name>
            <price>1374.01</price>
```

```
        </Product>
        <Product>
          <name>EMEND</name>
          <price>1365.51</price>
        </Product>
...
</Users>
```

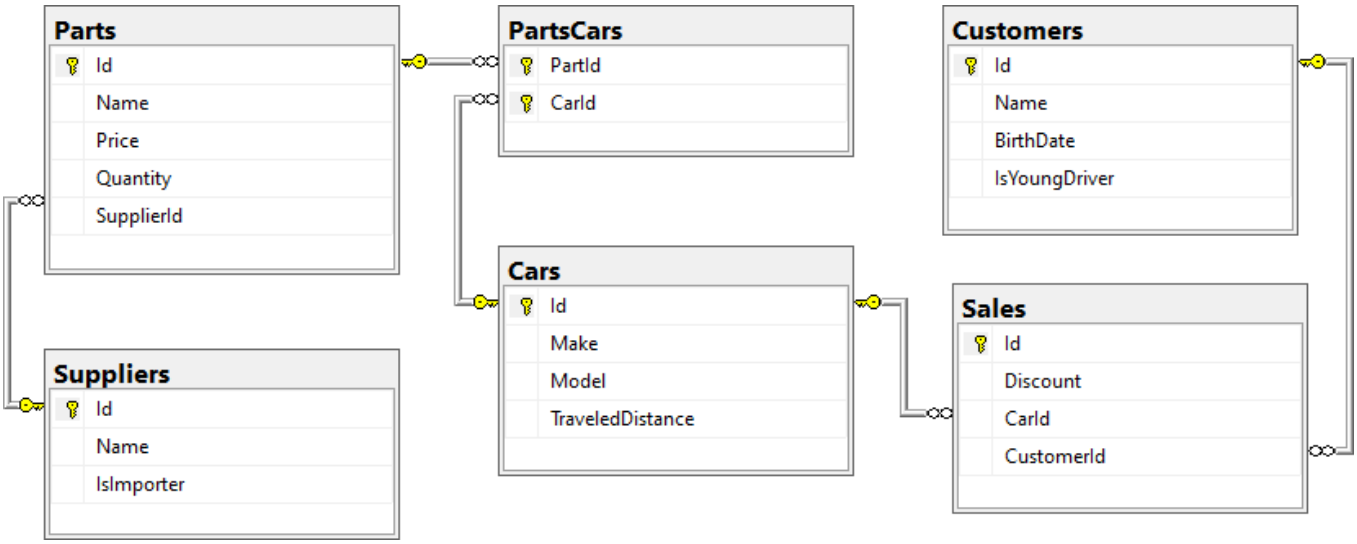
Car Dealer

1. Setup Database

A car dealer needs information about cars, their parts, parts suppliers, customers and sales.

- **Cars** have **Make, Model**, **TraveledDistance** in kilometers.
- **Parts** have **Name**, **Price** and **Quantity.**
- **Supplier** has a **Name** and info whether they supply **imported parts.**
- **Customer** has a **Name**, **BirthDate** and info whether they are **a young driver** (young driver is a driver that has **less than 2 years of experience**. Those customers get **an additional 5% off** for the sale.).
- **Sale** has **a Car**, **Customer** and a **Discount percentage.**

A **Price of a Car** is formed by **the total price** of **its Parts**.



- A **Car** has **many Parts** and **one Part** can be placed **in many Cars.**
- **One Supplier** can supply **many Parts** and each **Part** can be delivered by **only one Supplier.**
- In **one Sale**, only **one Car** can be sold to only one **Customer**.
- **A Customer** can buy **many Cars.**

2. Import Data

Import data from the provided files ("**suppliers.xml**", "**parts.xml**", "**cars.xml**", "**customers.xml**").

Query 9. Import Suppliers

NOTE: You will need method **public static string ImportSuppliers(CarDealerContext context, string inputXml)** and **public StartUp** class.

Import the suppliers from the provided file "**suppliers.xml**".

Your method should return a string with the following message:
`$"Successfully imported {suppliers.Count}";`

Query 10. Import Parts

NOTE: You will need method **`public static string ImportParts(CarDealerContext context, string inputXml)`** and **`public StartUp`** class.

Import the parts from the provided file "**`parts.xml`**". If the **`supplierId`** doesn't exist, skip the record.

Your method should return a string with the message:
`$"Successfully imported {parts.Count}";`

Query 11. Import Cars

NOTE: You will need method **`public static string ImportCars(CarDealerContext context, string inputXml)`** and **`public StartUp`** class.

Import the cars from the provided file "**`cars.xml`**". Select unique car part ids. If the **`partId`** doesn't exist, skip the **`Part`** record.

Your method should return a string with the following message:
`$"Successfully imported {cars.Count}";`

Query 12. Import Customers

NOTE: You will need method **`public static string ImportCustomers(CarDealerContext context, string inputXml)`** and **`public StartUp`** class.

Import the customers from the provided file "**`customers.xml`**".

Your method should return a string with the following message:
`$"Successfully imported {customers.Count}";`

Query 13. Import Sales

NOTE: You will need method **`public static string ImportSales(CarDealerContext context, string inputXml)`** and **`public StartUp`** class.

Import the sales from the provided file "**`sales.xml`**". If car doesn't exist, skip whole entity.

Your method should return a string with the following message:
`$"Successfully imported {sales.Count}";`

3. Query and Export Data

Write the below-described queries and **export** the returned data to the specified **format**. Make sure that Entity Framework generates only a **single query** for each task.

Query 14. Export Cars With Distance

NOTE: You will need method **`public static string GetCarsWithDistance(CarDealerContext context)`** and **`public StartUp`** class.

Get all **cars** with a distance of more **than** 2,000,000. Order them by make, then by model alphabetically. Take top 10 records.

Return the list of suppliers to **XML** in the format provided below.

cars.xml

```
<?xml version="1.0" encoding="utf-16"?>
<cars>
  <car>
    <make>BMW</make>
    <model>1M Coupe</model>
    <traveled-distance>39826890</traveled-distance>
  </car>
  <car>
    <make>BMW</make>
    <model>E67</model>
    <traveled-distance>476830509</traveled-distance>
  </car>
  <car>
    <make>BMW</make>
    <model>E88</model>
    <traveled-distance>27453411</traveled-distance>
  </car>
  ...
</cars>
```

Query 15. Export Cars from Make BMW

NOTE: You will need method **public static string GetCarsFromMakeBmw(CarDealerContext context)** and **public StartUp** class.

Get all **cars** from make **BMW** and **order them by model alphabetically** and by **traveled distance descending**.

Return the list of suppliers **to XML** in the format provided below.

bmw-cars.xml

```
<cars>
  <car id="26" model="1M Coupe" traveled-distance="39826890" />
  <car id="28" model="E67" traveled-distance="476830509" />
  <car id="24" model="E88" traveled-distance="27453411" />
  ...
</cars>
```

Query 16. Export Local Suppliers

NOTE: You will need method **public static string GetLocalSuppliers(CarDealerContext context)** and **public StartUp** class.

Get all **suppliers** that **do not import parts from abroad**. Get their **id**, **name** and **the number of parts they can offer to supply**.

Return the list of suppliers **to XML** in the format provided below.

local-suppliers.xml

```
<?xml version="1.0" encoding="utf-16"?>
<suppliers>
```

```
  <supplier id="2" name="Agway Inc." parts-count="3" />
  <supplier id="4" name="Airgas, Inc." parts-count="2" />
  ...
</suppliers>
```

Query 17. Export Cars with Their List of Parts

NOTE: You will need method **public static string GetCarsWithTheirListOfParts(CarDealerContext context)** and **public StartUp** class.

Get all **cars along with their list of parts**. For the **car** get only **make, model** and **traveled distance** and for the **parts** get only **name** and **price** and sort all parts by price (descending). Sort all cars by traveled **distance** (descending) and then by the **model** (ascending). Select top 5 records.

Return the list of suppliers **to XML** in the format provided below.

cars-and-parts.xml

```
<?xml version="1.0" encoding="utf-16"?>
<cars>
  <car make="Opel" model="Astra" traveled-distance="516628215">
    <parts>
      <part name="Tappet" price="300.29" />
      <part name="Front Left Side Door Glass" price="100.92" />
      <part name="Fan belt" price="10.99" />
    </parts>
  </car>
  ...
</cars>
```

Query 18. Export Total Sales by Customer

NOTE: You will need method **public static string GetTotalSalesByCustomer(CarDealerContext context)** and **public StartUp** class.

Get all **customers** that have bought **at least 1 car** and get their **names**, **bought cars count** and **total spent money** on cars. **Order** the result list **by total spent money** (descending). **Don't forget that young drivers get a discount**!

Return the list of suppliers **to XML** in the format provided below.

customers-total-sales.xml

```
<?xml version="1.0" encoding="utf-16"?>
<customers>
  <customer full-name="Emmitt Benally" bought-cars="1" spent-money="5044.10" />
  <customer full-name="Garret Capron" bought-cars="1" spent-money="3327.10" />
  <customer full-name="Cinthia Lasala" bought-cars="1" spent-money="3173.09" />
 ...
</customers>
```

Query 19. Export Sales with Applied Discount

NOTE: You will need method **public static string GetSalesWithAppliedDiscount(CarDealerContext context)** and **public StartUp** class.

Get all **sales** with information about the **car**, **customer** and **price** of the sale **with and without discount**. **Ignore any young driver discount in this calculation!**

- **Retrieve Sales Data**
  - Fetch all sales including details about the car (make, model, and traveled distance), customer, and the prices
- **Discount Handling**
  - Convert the discount value **from decimal** (as defined in the data model) **to an integer percentage**
- **Price Calculation**
  - **Price:** Should be represented as a **decimal value**
  - **Price with 5% Discount:** Convert the discount to a decimal and calculate the price after applying the discount. The result should be a decimal value rounded to four decimal places
- Formatting Requirements
  - **Price and Price with Discount:** Ensure these values are rounded to **four decimal places** when exporting

Return the list of suppliers **to XML** in the format provided below.

sales-discounts.xml

```
<?xml version="1.0" encoding="utf-16"?>
<sales>
  <sale>
    <car make="Opel" model="Omega" traveled-distance="109910837" />
    <discount>30</discount>
    <customer-name>Zada Attwoood</customer-name>
    <price>330.97</price>
    <price-with-discount>231.68</price-with-discount>
  </sale>
  ...
</sales>
```